

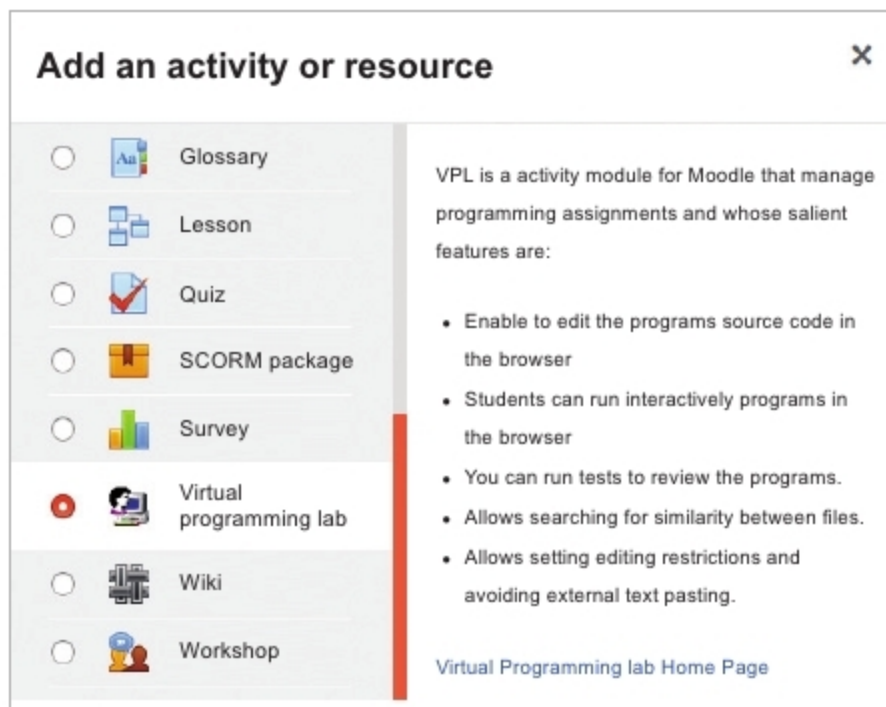


Figure 5: Adding a VPL activity

better, because you will be able to make the evaluation more rigorous. You also need to provide submission and grading related parameters for the test, like how many submissions are allowed, the deadline for submissions, etc. Now the test is ready; so let us attend it. Also remember that just like CodeRunner, any number of questions can be added in a VPL activity. Participants can log in as students or guest users to attempt this, depending on the privileges given to them. After logging in, the participants can either upload their program or use the editor, like in CodeRunner. Figure 6 shows the program execution environment of VPL.

For this article, we have used a Python script called *a.py*, which finds the square of a given number. This program can be executed directly from the VPL environment. The jail server automatically detects *a.py* as a Python script by the extension *py*. On execution, the jail server will automatically check the script and give the result shown in Figure 7. From the figure, it is clear that there was only a single test case provided for evaluating the script and it has passed that test. So, in this manner, VPL can be used to create and conduct online programming tests. Only the bare minimum features of VPL are discussed here; for a more detailed discussion, please refer to the Moodle plugins directory.

Additional features of VPL

Earlier we have seen how VPL allows us to check for plagiarism. We also saw how the jail server will automatically assign compilers to test programs written in different languages. But plagiarism checking and automatic compiler selection are not the only attractive features provided by VPL. Just like in CodeRunner, VPL also allows users to edit the program source code in the browser. Another feature that is available with both CodeRunner and VPL is the ability to run programs interactively in the browser. One major difference between CodeRunner and VPL is that VPL allows you to upload program files written offline for verification.



Figure 6: VPL environment for program execution

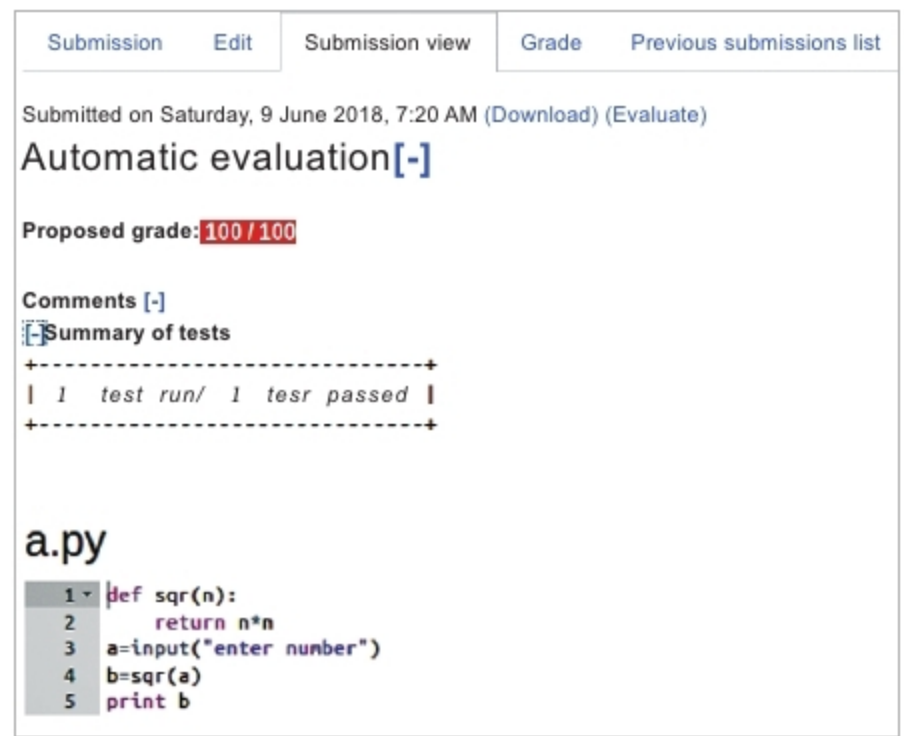


Figure 7: Automatic evaluation by VPL

This is a very useful feature when the assignment involves large programs that might take a lot of time to complete. Another advantage of VPL is that it allows you to run tests to review the programs rather than the final submission for grading. In CodeRunner, incorrect submissions may lead to a reduction in the score for that program. VPL also allows users to set editing restrictions on the programming environment like disabling external text pasting. These features and the plagiarism-checking capability of VPL make it a powerful tool in the hands of online technology content providers.

We hope that the article will create a general awareness regarding Moodle plugins and motivate a lot of people to explore the hundreds of Moodle plugins available online. Exploring and adopting Moodle and its plugins will lead to a better teaching-learning process, and education will then become edutainment. **END** 🐧

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